

Filters

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On behalf of Fink

Get used to the data

- Get data from the data transfer

  Github with help to read the data: https://github.com/emilleishida/Fink_IUCAA2026

- Read and apply cuts to the data

- Minimalist data download:
  https://github.com/emilleishida/Fink_IUCAA2026/blob/main/data/download_data.txt

Test and submit

https://doc.lsst.fink-broker.org/developers/filter_tutorial/

- For fink-filters github repo: https://github.com/astrolabsoftware/fink-filters
- Clone in your fork local working directory
- Get the docker image (this is big!)

```
>> docker pull
gitlab-registry.in2p3.fr/astrolabsoftware/fink/fink-deps-sentinel-rubin:latest
```

- Go to </path/to/fink-filters>
- Mount the docker image

```
>> docker run -t -i --rm -v $PWD:/home/libs/fink-filters \
gitlab-registry.in2p3.fr/astrolabsoftware/fink/fink-deps-sentinel-rubin:latest
bash
```

- Uninstall the docker version of fink-filters and install your local one

```
>> cd /home/libs/fink-filters/ && pip uninstall -y fink_filters && pip install -e .
```



Test and deploy

https://doc.lsst.fink-broker.org/developers/filter_tutorial/

- Choose a filter that will be your baseline and duplicate one of the directories in ~/fink-filters/fink_filters/rubin/livestream
- Duplicate the folder of the chosen filter, change the folder name and the name of the main function on the file filter.py to your name choice. It should start with `filter_`. Here, I will call mine `filter_iucaa`
- Change the name of the of the filter also in the tests (docstring inside the function).
- Without making any change to the function itself, test if the setup is working
  Inside the docker, at /home/libs/fink-filters/ do
  ```
  >> ./run_tests.sh --single_module \
  fink_filters/rubin/livestream/filter_iucaa/filter.py
  ```
- You should see a summary of the test of all filters. If everything went correct, at your filter you will see something like

```
fink_filters/rubin/livestream/filter_iucaa/__init__.py                    0      0   100%
```



Test and deploy

- Make the changes you want inside the `iucaa_candidates` function. Remember to also update the tests.
- If you need different data to test, get it from the data transfer service. Remember to keep the tests dataset minimum.
- Once you are finished with the modifications, test again to make sure they are properly integrated to the broker system

```
>> ./run_tests.sh --single_module \
fink_filters/rubin/livestream/filter_iucaa/filter.py
```

- Open a pull request (PR) from your fork to the main Fink repository
- Once your PR is approved you will be able to access the results of your filter from the kafka stream or the science portal



Once your filter is merged

https://doc.lsst.fink-broker.org/services/bots/

- Create a new telegram bot
  https://core.telegram.org/bots/features#creating-a-new-bot

  Search for BotFather
  Type /newbot --- follow instructions
  Create a new channel, add your bot and make it admin